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Mean Curvature Flow of Surface in 4-Manifolds[1]

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1. INTRODUCTION

Deforming submanifolds via various geometric parabolic flows has been a powerful method in differential geometry. In [B], Brakke introduced the motion of a submanifold moving by its mean curvature in arbitrary codimension and constructed a generalized varifold solution for all time. For the classical solution of the mean curvature flow, to our knowledge most work has been on hypersurfaces. Huisken showed in [H2, H3] that if the initial hypersurface is compact and uniformly convex in a complete manifold with bounded geometry then it converges to a single point under the mean curvature flow in a finite time and the normalized flow (area is fixed) converges to a sphere of that area in infinite (rescaled) time. As time evolves, the mean curvature flow may develop singularities which can be

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classified as Type I and Type II according to the blow-up rate of the second fundamental form with respect to time t. It is shown in [H1] that after appropriate rescaling near the Type I singularities the hypersurfaces Σ_t approach a self-similar solution of the mean curvature flow. The existence, uniqueness, and regularity of a weak solution of the mean curvature flow (the so-called viscosity solution) were studied by Chen *et al.* [CGG], Evans and Spruck [ES], White [W], and others. There is a fairly large amount of work on parabolic equations for plane curves or curves in Riemann surfaces moving by different geometric flows.

In this article, we are mainly interested in the mean curvature flow of compact real two dimensional symplectic surfaces in a Kähler–Einstein surface. The aim is to use the mean curvature flow method to produce holomorphic curves from a given initial symplectic surface. This problem is closely related to the well-known symplectic isotopy problem. Recall that two embedded symplectic surfaces are symplectic isotopy if they can be connected by a smooth family of embedded symplectic surfaces. The symplectic isotopy problem asks how many connected embedded symplectic surfaces represent a given 2-dimensional homology class in a simply connected symplectic 4-manifold, up to isotopy. Siebert and Tian conjectured (cf. [ST]) that for an algebraic surface with positive first Chern class the isotopy class of embedded symplectic surfaces in a fixed homology class is unique.

The organization of this paper is as follows. We first derive the parabolic differential equations for the second fundamental form and the mean curvature vector for submanifolds moving by the mean curvature flow in arbitrary codimension, which does not seem to be easily found in literature except in the codimension one case. The Laplacian of the second fundamental form was computed by Simons [Sim] a long time ago. Then we concentrate on surfaces in 4-manifolds. In particular, we assume that the initial surface is compact and symplectic in a Kähler surface, which means that the Kähler angle α at every point of the surface lies in $[0, \frac{\pi}{2})$. We recover a result in [CT2] which states that the symplectic property is preserved under the mean curvature flow in a Kähler–Einstein surface as long as the smooth solution exists. The monotonicity formulae for the area functional in the geometric measure theory and for the Dirichlet integral in harmonic mapping theory have been useful tools to provide regularity results. We establish a monotonicity formula for $\int_{\Sigma_t} \frac{1}{\cos\alpha}\, \rho(F, t)\, \phi\, d\mu_t$, where $F\colon \Sigma_0 \times [0, T) \to M$ is the mean curvature flow, ϕ is a cut-off function supported in a small geodesic ball in M which is contained in a single coordinate chart, and ρ is the backward heat kernel on R^4. The idea of using various backward heat kernels (on the domain or the entire target) appeared in the earlier work of other people on the mean curvature flow, the harmonic map heat flow and the Yang–Mills heat flow and so on (cf. [CS, Ha, H1, St], etc.). Using our monotonicity formula and a blowing up argument near singularities, we prove the main result of this article: if the initial surface is compact and symplectic in a Kähler–Einstein surface with nonnegative scalar curvature, then the mean curvature flow has no Type I singularities. We should point out that if the mean curvature flow has a global solution, then $F(\Sigma, t)$ will converge to holomorphic curves.

We take summation for repeated indices throughout the paper.

2. BASIC FORMULAE FOR MEAN CURVATURE FLOW IN ARBITRARY DIMENSION AND CO-DIMENSION

In this section, we derive the equations for some geometric quantities. The results in this section hold for the mean curvature flow in arbitrary dimension and codimension.

We consider the mean curvature flow from a closed P-dimensional manifold in a Q-dimensional Riemannian manifold M with a Riemannian metric. Given an embedding $F_0\colon \Sigma \to M$, we consider a one-parameter family of smooth maps $F_t = F(\cdot, t)\colon \Sigma \to M$ with corresponding images $\Sigma_t = F_t(\Sigma)$ are embedded submanifolds in M and F satisfies the mean curvature flow equation:

$$\begin{cases} \dfrac{d}{dt} F(x, t) = \mathbf{H}(x, t) \\ F(x, 0) = F_0(x). \end{cases} \tag{1}$$

Here $\mathbf{H}(x, t)$ is the mean curvature vector of Σ_t at $F(x, t)$ in M. Denote by $\mathbf{A}$ the second fundamental form of Σ_t in M and the Riemannian metric on M by $\langle \cdot, \cdot \rangle$. In a normal coordinates around a point in Σ, the induced metric on Σ_t from $\langle \cdot, \cdot \rangle$ is given by

$$g_{ij} = \langle \partial_i F, \partial_j F \rangle,$$

where ∂_i $(i = 1, ..., P)$ are the partial derivatives with respect to the local coordinates. In the sequel, we denote by Δ and ∇ the Laplace operator and covariant derivative for the induced metric on Σ_t respectively. We choose a local field of orthonormal frames $e_1, ..., e_P, v_1, ..., v_{Q-P}$ of M along Σ_t such that $e_1, ..., e_P$ are tangent vectors of Σ_t and $v_1, ..., v_{Q-P}$ are in the normal bundle over Σ_t. We can write

$$\mathbf{A} = A^\alpha v_\alpha$$

$$\mathbf{H} = -H^\alpha v_\alpha.$$

Let $A^\alpha = (h^\alpha_{ij})$ where (h^α_{ij}) is a matrix, by the Weingarten equation (cf. [Sp]), we have

$$h^\alpha_{ij} = \langle \partial_i v_\alpha, \partial_j F \rangle = \langle \partial_j v_\alpha, \partial_i F \rangle = h^\alpha_{ji}.$$

The trace and the norm of the second fundamental form of Σ_t in M are

$$H^\alpha = g^{ij} h^\alpha_{ij} = h^\alpha_{ii}$$

$$|\mathbf{A}|^2 = \sum_\alpha |A^\alpha|^2 = g^{ij} g^{kl} h^\alpha_{ik} h^\alpha_{jl} = h^\alpha_{ik} h^\alpha_{ik}.$$

We first derive the evolution equation of the induced metric.

LEMMA 2.1. *Under the mean curvature flow, the induced metric evolves as*

$$\frac{\partial g_{ij}}{\partial t} = -2H^\alpha h^\alpha_{ij}.$$

Proof. Using normal coordinate systems at x in Σ and at $F(x)$ in M, we have

$$\begin{aligned}\frac{\partial g_{ij}}{\partial t} &= \frac{\partial}{\partial t} \langle \partial_i F, \partial_j F \rangle \\ &= -\langle \partial_i (H^\alpha v_\alpha), \partial_j F \rangle - \langle \partial_j (H^\alpha v_\alpha), \partial_i F \rangle,\end{aligned}$$

where ∂_i $(i = 1, ..., P)$ are the partial derivatives with respect to the local coordinates. By the Gauss–Weingarten equations (cf. [H3, (5); Sp]), we have

$$\partial_i v_\alpha = h^\alpha_{il} g^{lk}\, \partial_k F + C^\beta_{i\alpha} v_\beta.$$

It is clear that $C^\beta_{i\alpha} = -C^\alpha_{i\beta}$. So, we get

$$\frac{\partial g_{ij}}{\partial t} = -2H^\alpha h^\alpha_{ij}.$$

This proves the lemma. Q.E.D.

We then derive the evolution equation of the frame on the normal bundle.

LEMMA 2.2. *We write $\langle \partial v_\alpha / \partial t, v_\beta \rangle = b^\beta_\alpha$, then $b^\beta_\alpha = -b^\alpha_\beta$ and*

$$\frac{\partial v_\alpha}{\partial t} = \nabla H^\alpha + H^\gamma C^\alpha_{i\gamma} e_i + b^\beta_\alpha v_\beta.$$

Proof. Differentiating the identity $\langle v_\alpha, v_\beta\rangle = \delta_{\alpha\beta}$ yields the skew-symmetry of b^α_β. Using normal coordinate systems at x in Σ and at $F(x)$ in M, then translating the identity into normal frames, we have

$$\begin{aligned}\frac{\partial v_\alpha}{\partial t} &= \left\langle \frac{\partial v_\alpha}{\partial t}, \partial_i F\right\rangle g^{ij}\,\partial_j F + b^\beta_\alpha v_\beta \\ &= -\left\langle v_\alpha, \partial_i\left(\frac{\partial F}{\partial t}\right)\right\rangle g^{ij}\,\partial_j F + b^\beta_\alpha v_\beta \\ &= \langle v_\alpha, \partial_i(H^\alpha v_\alpha)\rangle\, g^{ij}\,\partial_j F + b^\beta_\alpha v_\beta \\ &= \nabla H^\alpha + H^\gamma \langle v_\alpha, C^\beta_{i\gamma} v_\beta\rangle\, e_i + b^\beta_\alpha v_\beta \\ &= \nabla H^\alpha + H^\gamma C^\alpha_{i\gamma} e_i + b^\beta_\alpha v_\beta\end{aligned}$$

which concludes the proof of the lemma. Q.E.D.

Note that for hypersurfaces the terms b^α_β are identically zero.

LEMMA 2.3. *Assume that the Christoffel symbols Γ^k_{ij} of the Levi–Civita connection of the induced metric are zero at a point $p \in \Sigma_t$. Then at p*

$$\begin{aligned}\frac{\partial}{\partial t} h^\alpha_{ij} = {}& \nabla_i\nabla_j H^\alpha - H^\gamma h^\gamma_{jl} h^\alpha_{il} + H^\gamma C^\beta_{j\gamma} C^\alpha_{i\beta} + H^\gamma \nabla_i C^\alpha_{j\gamma} \\ & + \nabla_j H^\beta C^\alpha_{i\beta} + \nabla_i H^\beta C^\alpha_{j\beta} - h^\beta_{ij} b^\alpha_\beta + H^\beta R_{\alpha j\beta i}.\end{aligned}$$

Proof. We will first calculate in local normal coordinates and then translate it into normal frames as before. By the Gauss–Weingarten equations (cf. [H3, (5); Sp]), we have

$$h^\alpha_{ij} = -\langle \partial^2_{ij} F, v_\alpha\rangle + \Gamma^\alpha_{ij}.$$

Then we have

$$\begin{aligned}\frac{\partial}{\partial t} h^\alpha_{ij} &= -\frac{\partial}{\partial t}\langle \partial^2_{ij} F, v_\alpha\rangle - H^\beta\,\partial_\beta \Gamma^\alpha_{ij} \\ &= \langle \partial^2_{ij}(H^\alpha v_\alpha), v_\alpha\rangle - \langle \partial^2_{ij} F, \nabla H^\alpha + H^\gamma C^\alpha_{i\gamma} e_i + b^\beta_\alpha v_\beta\rangle - H^\beta\,\partial_\beta \Gamma^\alpha_{ij}.\end{aligned}$$

By the Gauss formula (cf. [Sp]) and the Weingarten equations, we have that at the point p

$$\partial^2_{ij} F = -h^\alpha_{ij} v_\alpha$$

and

$$\partial_i v_\alpha = h^\alpha_{il} g^{lm}\,\partial_m F + C^\beta_{i\alpha} v_\beta,$$

we obtain

$$\frac{\partial}{\partial t} h^{\alpha}_{ij} = \nabla^2_{ij} H^{\alpha} - H^{\gamma} h^{\gamma}_{jl} h^{\alpha}_{il} + H^{\gamma} C^{\beta}_{j\gamma} C^{\alpha}_{i\beta} - H^{\beta}\, \partial_{\beta} \Gamma^{\alpha}_{ij} + H^{\beta}\, \partial_i \Gamma^{\alpha}_{j\beta}$$
$$+ \langle \nabla_j H^{\gamma} C^{\beta}_{i\gamma} v_{\beta}, v_{\alpha} \rangle + \langle \nabla_i H^{\gamma} C^{\beta}_{j\alpha} v_{\beta}, v_{\gamma} \rangle + H^{\gamma} \nabla_i C^{\alpha}_{j\gamma} - h^{\beta}_{ij} b^{\alpha}_{\beta}.$$

This proves the lemma. Q.E.D.

LEMMA 2.4. *Assume that the Christoffel symbols of the Levi–Civita connection of the induced metric are zero at $p \in \Sigma_t$. Then at the point p we have*

$$\nabla_i \nabla_j H^{\alpha} = \Delta h^{\alpha}_{ij} + \nabla_l (h^{\beta}_{ij} C^{\alpha}_{l\beta}) - \nabla_l (h^{\beta}_{lj} C^{\alpha}_{i\beta}) + h^{\beta}_{il} h^{\beta}_{lm} h^{\alpha}_{mj} - H^{\beta} h^{\beta}_{im} h^{\alpha}_{mj}$$
$$+ h^{\beta}_{ij} h^{\beta}_{lm} h^{\alpha}_{ml} - h^{\beta}_{im} h^{\beta}_{lj} h^{\alpha}_{ml} + \nabla_i (h^{\beta}_{jl} C^{\alpha}_{l\beta}) - \nabla_i (H^{\beta} C^{\alpha}_{j\beta})$$
$$- \nabla_l R_{\alpha jil} - \nabla_i R_{\alpha ljl} + R_{illm} h^{\alpha}_{mj} + R_{iljm} h^{\alpha}_{ml} - R_{\alpha\beta il} h^{\beta}_{lj}.$$

Proof. Let K be the curvature tensor on Σ_t. By the Gauss equation and the Codazzi–Mainardi equations, we have the following identities at p,

$$\nabla_k h^{\alpha}_{mn} - \nabla_m h^{\alpha}_{kn} = h^{\beta}_{kn} C^{\alpha}_{m\beta} - h^{\beta}_{mn} C^{\alpha}_{k\beta} - R_{\alpha nmk}$$

and

$$K_{ijkl} = (h^{\beta}_{ik} h^{\beta}_{jl} - h^{\beta}_{il} h^{\beta}_{jk}) + R_{ijkl},$$

where R is the curvature tensor of M. We have

$$\begin{aligned}
\nabla_i \nabla_j H^{\alpha} &= \nabla_i \nabla_j h^{\alpha}_{ll} \\
&= \nabla_i (\nabla_l h^{\alpha}_{jl} + h^{\beta}_{jl} C^{\alpha}_{l\beta} - h^{\beta}_{ll} C^{\alpha}_{j\beta} - R_{\alpha ljl}) \\
&= \nabla_i \nabla_l h^{\alpha}_{lj} + \nabla_i (h^{\beta}_{jl} C^{\alpha}_{l\beta}) - \nabla_i (h^{\beta}_{ll} C^{\alpha}_{j\beta}) - \nabla_i R_{\alpha ljl} \\
&= \nabla_l \nabla_i h^{\alpha}_{lj} + (h^{\beta}_{il} h^{\beta}_{lm} - h^{\beta}_{im} h^{\beta}_{ll})\, h^{\alpha}_{mj} + R_{illm} h^{\alpha}_{mj} \\
&\quad + (h^{\beta}_{ij} h^{\beta}_{lm} - h^{\beta}_{im} h^{\beta}_{lj})\, h^{\alpha}_{ml} + R_{iljm} h^{\alpha}_{ml} - R_{\alpha\beta il} h^{\beta}_{lj} \\
&\quad - \nabla_i R_{\alpha ljl} + \nabla_i (h^{\beta}_{jl} C^{\alpha}_{l\beta}) - \nabla_i (h^{\beta}_{ll} C^{\alpha}_{j\beta}) \\
&= \nabla_l (\nabla_l h^{\alpha}_{ij} + h^{\beta}_{ij} C^{\alpha}_{l\beta} - h^{\beta}_{lj} C^{\alpha}_{i\beta} - R_{\alpha jil}) + h^{\beta}_{il} h^{\beta}_{lm} h^{\alpha}_{mj} - H^{\beta} h^{\beta}_{im} h^{\alpha}_{mj} \\
&\quad + h^{\beta}_{ij} h^{\beta}_{lm} h^{\alpha}_{ml} - h^{\beta}_{im} h^{\beta}_{lj} h^{\alpha}_{ml} + \nabla_i (h^{\beta}_{jl} C^{\alpha}_{l\beta}) - \nabla_i (H^{\beta} C^{\alpha}_{j\beta}) \\
&\quad + R_{illm} h^{\alpha}_{mj} + R_{iljm} h^{\alpha}_{ml} - R_{\alpha\beta il} h^{\beta}_{lj} - \nabla_i R_{\alpha ljl} \\
&= \Delta h^{\alpha}_{ij} + \nabla_l (h^{\beta}_{ij} C^{\alpha}_{l\beta}) - \nabla_l (h^{\beta}_{lj} C^{\alpha}_{i\beta}) + h^{\beta}_{il} h^{\beta}_{lm} h^{\alpha}_{mj} - H^{\beta} h^{\beta}_{im} h^{\alpha}_{mj} \\
&\quad + h^{\beta}_{ij} h^{\beta}_{lm} h^{\alpha}_{ml} - h^{\beta}_{im} h^{\beta}_{lj} h^{\alpha}_{ml} + \nabla_i (h^{\beta}_{jl} C^{\alpha}_{l\beta}) - \nabla_i (H^{\beta} C^{\alpha}_{j\beta}) \\
&\quad - \nabla_l R_{\alpha jil} - \nabla_i R_{\alpha ljl} + R_{illm} h^{\alpha}_{mj} + R_{iljm} h^{\alpha}_{ml} - R_{\alpha\beta il} h^{\beta}_{lj}.
\end{aligned}$$

This proves the lemma. Q.E.D.

From the Codazzi–Mainardi equation, it follows that

$$(\nabla_j H^\beta)\, C^\alpha_{i\beta} - \nabla_l (h^\beta_{jl} C^\alpha_{i\beta}) = h^\gamma_{jl} C^\beta_{l\gamma} C^\alpha_{i\beta} - H^\gamma C^\beta_{j\gamma} C^\alpha_{i\beta} - R_{\beta llj} C^\alpha_{i\beta} - h^\beta_{jl} \nabla_l C^\alpha_{i\beta}.$$

Then Lemma 2.3 and Lemma 2.4 immediately imply

LEMMA 2.5. *For the mean curvature flow, the second fundamental form satisfies*

$$\begin{aligned}\left(\frac{\partial}{\partial t} - \Delta\right) h_{ij} &= \nabla_l (h^\beta_{ij} C^\alpha_{l\beta}) + h^\beta_{il} h^\beta_{lm} h^\alpha_{mj} \\ &\quad - H^\beta (h^\beta_{im} h^\alpha_{mj} + h^\beta_{jl} h^\alpha_{il}) + h^\beta_{ij} h^\beta_{lm} h^\alpha_{ml} - h^\beta_{im} h^\beta_{jl} h^\alpha_{ml} \\ &\quad + h^\beta_{jl} (\nabla_i C^\alpha_{l\beta} - \nabla_l C^\alpha_{i\beta}) + h^\gamma_{jl} C^\beta_{l\gamma} C^\alpha_{i\beta} - R_{\beta llj} C^\alpha_{i\beta} \\ &\quad - \nabla_l R_{\alpha jil} - \nabla_i R_{\alpha ljl} + R_{illm} h^\alpha_{mj} + R_{iljm} h^\alpha_{ml} - R_{\alpha\beta il} h^\beta_{lj} - h^\beta_{ij} b^\alpha_\beta.\end{aligned}$$

Now we prove the main result in this section.

PROPOSITION 2.6. *Suppose that F is the mean curvature flow. Then*

$$\begin{aligned}\left(\frac{\partial}{\partial t} - \Delta\right) |\mathbf{A}|^2 &= -2\, |\tilde{\nabla} \mathbf{A}|^2 + 2\, |\mathbf{A}|^4 - 2 h^\alpha_{ij} (\bar{\nabla}_l R_{\alpha jil} + \bar{\nabla}_i R_{\alpha ljl}) \\ &\quad + 2 h^\alpha_{ij} h^\beta_{ij} R_{\alpha l\beta l} + 4 h^\alpha_{ij} R_{illm} h^\alpha_{mj} + 4 h^\alpha_{ij} R_{iljm} h^\alpha_{ml}.\end{aligned}$$

and

$$\left(\frac{\partial}{\partial t} - \Delta\right) |\mathbf{H}|^2 = -2\, |\tilde{\nabla} \mathbf{H}|^2 + 2\, |\mathbf{A}|^2\, |\mathbf{H}|^2 + 2 H^\alpha H^\beta R_{l\alpha l\beta},$$

where $\tilde{\nabla}$ *is the covariant differentiation on* $\mathrm{Hom}(T\Sigma_t \times T\Sigma_t, Nor\Sigma_t)$ *determined by the covariant differentiation on* $T\Sigma_t$ *and D on the normal bundle, D is the normal connection for the embedding* $\Sigma_t \subset M$ *(cf. [Sp, p. 55]), and* $\bar{\nabla}$ *is the connection on* M.

Proof. By the skew-symmetry of b^α_β, it is clear that

$$h^\alpha_{ij} h^\beta_{ij} b^\alpha_\beta = 0.$$

Then using the previous lemmas, we have

$$
\begin{aligned}
\frac{\partial}{\partial t}|\mathbf{A}|^2 &= \frac{\partial}{\partial t}(g^{ik}g^{jl}h^\alpha_{ij}h^\alpha_{kl}) \\
&= 4H^\alpha h^\alpha_{ik}h^\beta_{ij}h^\beta_{kj} + 2h^\alpha_{ij}\frac{\partial}{\partial t}h^\alpha_{ij} \\
&= 2h^\alpha_{ij}(\Delta h^\alpha_{ij} + H^\gamma C^\beta_{j\gamma}C^\alpha_{i\beta} + H^\gamma \nabla_i C^\alpha_{j\gamma} + \nabla_j H^\beta C^\alpha_{i\beta} + \nabla_i H^\beta C^\alpha_{j\beta} - h^\beta_{ij}b^\alpha_\beta \\
&\quad + \nabla_l(h^\beta_{ij}C^\alpha_{l\beta}) - \nabla_l(h^\beta_{lj}C^\alpha_{i\beta}) + h^\beta_{il}h^\beta_{lm}h^\alpha_{mj} + h^\beta_{ij}h^\beta_{lm}h^\alpha_{ml} - h^\beta_{im}h^\beta_{lj}h^\alpha_{ml} \\
&\quad - \nabla_l R_{\alpha jil} - \nabla_i R_{\alpha ljl} + R_{illm}h^\alpha_{mj} + R_{iljm}h^\alpha_{ml} - R_{\alpha\beta il}h^\beta_{lj} + H^\beta R_{\alpha j\beta i} \\
&\quad + \nabla_i(h^\beta_{jl}C^\alpha_{l\beta}) - \nabla_i(H^\beta C^\alpha_{j\beta})) \\
&= 2h^\alpha_{ij}(\Delta h^\alpha_{ij} + H^\gamma C^\beta_{j\gamma}C^\alpha_{i\beta} + H^\gamma \nabla_i C^\alpha_{j\gamma} + \nabla_j H^\beta C^\alpha_{i\beta} + \nabla_i H^\beta C^\alpha_{j\beta} \\
&\quad + \nabla_l h^\beta_{ij}C^\alpha_{l\beta} + h^\beta_{ij}\nabla_l C^\alpha_{l\beta} - \nabla_l h^\beta_{lj}C^\alpha_{i\beta} - h^\beta_{lj}\nabla_l C^\alpha_{i\beta} \\
&\quad + \nabla_i h^\beta_{jl}C^\alpha_{l\beta} + h^\beta_{jl}\nabla_i C^\alpha_{l\beta} - \nabla_i H^\beta C^\alpha_{j\beta} - H^\beta \nabla_i C^\alpha_{j\beta} \\
&\quad + h^\beta_{il}h^\beta_{lm}h^\alpha_{mj} + h^\beta_{ij}h^\beta_{lm}h^\alpha_{ml} - h^\beta_{im}h^\beta_{lj}h^\alpha_{ml} \\
&\quad - \nabla_l R_{\alpha jil} - \nabla_i R_{\alpha ljl} + R_{illm}h^\alpha_{mj} + R_{iljm}h^\alpha_{ml} - R_{\alpha\beta il}h^\beta_{lj} + H^\beta R_{\alpha j\beta i}).
\end{aligned}
$$

Note that

$$
\begin{aligned}
\nabla_l h^\beta_{jl} &= \nabla_j h^\beta_{ll} + H^\gamma C^\beta_{j\gamma} - h^\gamma_{jl}C^\beta_{l\gamma} - R_{\beta ljl} \\
\nabla_i h^\beta_{lj} &= \nabla_l h^\beta_{ij} + h^\gamma_{ij}C^\beta_{l\gamma} - h^\gamma_{jl}C^\beta_{i\gamma} - R_{\beta jli} \\
\nabla_i C^\alpha_{l\beta} - C^\gamma_{i\beta}C^\alpha_{l\gamma} - \nabla_l C^\alpha_{i\beta} + C^\gamma_{l\beta}C^\alpha_{i\gamma} &= h^\alpha_{lk}h^\beta_{ki} - h^\alpha_{ik}h^\beta_{kl} + R_{il\alpha\beta}.
\end{aligned}
$$

We then obtain

$$
\begin{aligned}
\frac{\partial}{\partial t}|\mathbf{A}|^2 &= 2(h^\alpha_{ij}\Delta h^\alpha_{ij} - 2h^\alpha_{ij}\nabla_l h^\beta_{ij}C^\beta_{l\alpha} - (h^\alpha_{ij}C^\beta_{l\alpha})^2) + 2h^\alpha_{ij}h^\beta_{ij}h^\beta_{lm}h^\alpha_{ml} \\
&\quad + 2h^\alpha_{ij}(-\nabla_l R_{\alpha jil} - \nabla_i R_{\alpha ljl} + R_{illm}h^\alpha_{mj} + R_{iljm}h^\alpha_{ml} + H^\beta R_{\alpha j\beta i}). \qquad (2)
\end{aligned}
$$

Covariant differentiation of the curvature tensor leads to the following formula (cf. [SSY]),

$$\nabla_q R_{\alpha mnp} = \overline{\nabla}_q R_{\alpha mnp} - R_{\alpha m\beta p}h^\beta_{nq} - R_{\alpha mn\beta}h^\beta_{pq} + R_{smnp}h^\alpha_{sq},$$

and in turn we have

$$\nabla_l R_{\alpha jil} = \overline{\nabla}_l R_{\alpha jil} - R_{\alpha j\beta l}h^\beta_{il} - R_{\alpha ji\beta}h^\beta_{ll} + R_{mjil}h^\alpha_{ml},$$

and

$$\nabla_i R_{\alpha ljl} = \overline{\nabla}_i R_{\alpha ljl} - R_{\alpha l\beta l} h^{\beta}_{ij} - R_{\alpha lj\beta} h^{\beta}_{il} + R_{mljl} h^{\alpha}_{mi}.$$

It is clear that

$$\Delta\,|\mathbf{A}|^2 = 2h^{\alpha}_{ij}\,\Delta h^{\alpha}_{ij} + 2\nabla_l h^{\alpha}_{ij} \nabla_l h^{\alpha}_{ij}.$$

Substituting the last three identities into (2), we get the first identity in the proposition. The second one can be proved by a similar argument. Q.E.D.

3. MOVING SYMPLECTIC CURVES IN KÄHLER–EINSTEIN SURFACES

In this section, we assume that M is a Kähler–Einstein surface. Let ω be the symplectic form on M and let J be a complex structure compatible with ω in the sense that for any tangent vectors U, V in TM, we have

$$\omega(JU, JV) = \omega(U, V).$$

The Riemannian metric $\langle\ ,\ \rangle$ on M is defined by

$$\langle U, V\rangle = \omega(U, JV).$$

Let J_{Σ_t} be an almost complex structure in a tubular neighborhood of Σ_t on M with

$$\begin{cases} J_{\Sigma_t} e_1 = & e_2 \\ J_{\Sigma_t} e_2 = & -e_1 \\ J_{\Sigma_t} v_1 = & v_2 \\ J_{\Sigma_t} v_2 = & -v_1. \end{cases} \tag{3}$$

It is not difficult to check that

$$(\overline{\nabla}_i J_{\Sigma_t})\, e_1 = (h^2_{i1} + h^1_{i2})\, v_1 + (h^2_{i2} - h^1_{i1})\, v_2$$

and

$$(\overline{\nabla}_i J_{\Sigma_t})\, e_2 = (h^2_{i1} + h^1_{i2})\, v_2 + (h^2_{i2} - h^1_{i1})\, v_1.$$

So we have

$$|\overline{\nabla} J_{\Sigma_t}|^2 = |h^2_{11} + h^1_{12}|^2 + |h^2_{21} + h^1_{22}|^2 + |h^2_{12} - h^1_{11}|^2 + |h^2_{22} - h^1_{21}|^2.$$

A simple computation shows the relation between $|\bar{\nabla} J_{\Sigma_t}|^2$ and $|\mathbf{H}|^2$ which we state as follows (cf. [CT]).

LEMMA 3.1. *We have*

$$|\bar{\nabla} J_{\Sigma_t}|^2 = \tfrac{1}{2}\,|\mathbf{H}|^2 + \tfrac{1}{2}\,(((h_{11}^1 + h_{22}^1) + 2(h_{12}^2 - h_{22}^1))^2 + (h_{11}^2 + h_{22}^2 + 2(h_{21}^1 - h_{11}^2))^2).$$

The contraction of the Kähler 2-form ω with the area element of Σ_t determines the Kähler angle α of Σ_t in M. Using the orthonormal frame e_1, e_2, we have

$$\cos\alpha = \langle e_1 \wedge e_2, \omega\rangle.$$

The function α is continuous everywhere on Σ_t and smooth possibly except at the complex or anti-complex points, i.e., where $\alpha = 0, \pi$. Whenever α is smooth, we have

PROPOSITION 3.2. *Assume that M is a Kähler–Einstein surface with scalar curvature R. Let α be the Kähler angle of the surface Σ_t which evolves by the mean curvature flow. Then*

$$\left(\frac{\partial}{\partial t} - \Delta\right)\cos\alpha = |\bar{\nabla} J_{\Sigma_t}|^2 \cos\alpha + R \sin^2\alpha \cos\alpha.$$

Remark. The above equation is a simplified version of the evolution equation on $\cos\alpha$ in [CT2]. This simplified version was pointed out to us by G. Tian.

Proof. We may choose the local orthonormal frame $\{e_1, e_2, e_3, e_4\}$ on M along Σ_t so that the symplectic form ω takes the following form (cf. [CT, CW]),

$$\omega = \cos\alpha u_1 \wedge u_2 + \cos\alpha u_3 \wedge u_4 + \sin\alpha u_1 \wedge u_3 - \sin\alpha u_2 \wedge u_4, \tag{4}$$

where $\{u_1, u_2, u_3, u_4\}$ is the dual frame of $\{e_1, e_2, e_3, e_4\}$. Then along the surface Σ_t the complex structure on M takes the form

$$J = \begin{pmatrix} 0 & \cos\alpha & \sin\alpha & 0 \\ -\cos\alpha & 0 & 0 & -\sin\alpha \\ -\sin\alpha & 0 & 0 & \cos\alpha \\ 0 & \sin\alpha & -\cos\alpha & 0 \end{pmatrix} \tag{5}$$

and we use J^i_j to denote the coordinate components of the type (1,1) tensor J. We set

$$v=\cos\alpha=\langle e_1\wedge e_2,\omega\rangle.$$

From Lemma 2.2, we have

$$\begin{aligned}\frac{\partial}{\partial t}e_i&=\left\langle\frac{\partial}{\partial t}e_i,e_j\right\rangle e_j+\left\langle\frac{\partial}{\partial t}e_i,v_\alpha\right\rangle v_\alpha\\&=\left\langle\frac{\partial}{\partial t}e_i,e_j\right\rangle e_j-\left\langle e_i,\frac{\partial}{\partial t}v_\alpha\right\rangle v_\alpha\\&=\left\langle\frac{\partial}{\partial t}e_i,e_j\right\rangle e_j-\nabla_i H^\alpha v_\alpha-H^\gamma C^\alpha_{i\gamma}v_\alpha.\end{aligned}$$

Since ω is independent of t, it follows

$$\begin{aligned}\frac{\partial}{\partial t}v&=\left\langle\frac{\partial}{\partial t}e_1\wedge e_2,\omega\right\rangle+\left\langle e_1\wedge\frac{\partial}{\partial t}e_2,\omega\right\rangle\\&=-(\nabla_1 H^\alpha+H^\gamma C^\alpha_{1\gamma})\langle v_\alpha\wedge e_2,\omega\rangle\\&\quad-(\nabla_2 H^\alpha+H^\gamma C^\alpha_{2\gamma})\langle e_1\wedge v_\alpha,\omega\rangle.\end{aligned}$$

For convenience of summing over indices, we denote v_1 by e_3 and v_2 by e_4. For the covariant derivatives of the orthonormal frame e_1, e_2, e_3, e_4 in M, we set

$$\bar\nabla_{e_i}e_j=\Gamma^k_{ij}e_k.$$

That $\bar\nabla$ is torsion-free and its connection matrix is skew-symmetric in the orthonormal frame $\{e_1,\dots,e_4\}$ asserts

$$\Gamma^k_{ij}=\Gamma^k_{ji}\qquad\text{and}\qquad\Gamma^k_{ij}=-\Gamma^j_{ik}.$$

From the Gauss equation

$$\nabla_i e_j=-(h^\alpha_{ij}-\Gamma^\alpha_{ij})\,v_\alpha+\Gamma^k_{ij}e_k,$$

we see that the function v satisfies

$$\begin{aligned}\nabla_1 v&=-((h^\alpha_{11}-\Gamma^{\alpha+2}_{11})\langle v_\alpha\wedge e_2,\omega\rangle+(h^\alpha_{12}-\Gamma^{\alpha+2}_{12})\langle e_1\wedge v_\alpha,\omega\rangle),\\\nabla_2 v&=-((h^\alpha_{21}-\Gamma^{\alpha+2}_{21})\langle v_\alpha\wedge e_2,\omega\rangle+(h^\alpha_{22}-\Gamma^{\alpha+2}_{22})\langle e_1\wedge v_\alpha,\omega\rangle).\end{aligned}$$

For the sake of simplicity, we assume that $\Gamma^k_{ij}=0$ at the point under consideration, for $i, j, k=1, ..., 4$. Then

$$\begin{aligned}
\nabla_1^2 v = & -\nabla_1(h^\alpha_{11}-\Gamma^{\alpha+2}_{11})\langle v_\alpha\wedge e_2, \omega\rangle - h^\alpha_{11}\langle\nabla_1 v_\alpha\wedge e_2, \omega\rangle + h^\alpha_{11}h^\beta_{12}\langle v_\alpha\wedge v_\beta, \omega\rangle\\
& -\nabla_1(h^\alpha_{12}-\Gamma^\alpha_{12})\langle e_1\wedge v_\alpha, \omega\rangle - h^\alpha_{12}\langle e_1\wedge\nabla_1 v_\alpha, \omega\rangle + h^\alpha_{11}h^\beta_{12}\langle v_\alpha\wedge v_\beta, \omega\rangle\\
= & -\nabla_1 h^\alpha_{11}\langle v_\alpha\wedge e_2, \omega\rangle - (h^\alpha_{11})^2\langle e_1\wedge e_2, \omega\rangle - h^\gamma_{11}C^\alpha_{1\gamma}\langle v_\alpha\wedge e_2, \omega\rangle\\
& -\nabla_1 h^\alpha_{12}\langle e_1\wedge v_\alpha, \omega\rangle - (h^\alpha_{12})^2\langle e_1\wedge e_2, \omega\rangle - h^\gamma_{12}C^\alpha_{1\gamma}\langle e_1\wedge v_\alpha, \omega\rangle\\
& +2h^\alpha_{11}h^\beta_{12}\langle v_\alpha\wedge v_\beta, \omega\rangle + \nabla_1\Gamma^{\alpha+2}_{11}\langle v_\alpha\wedge e_2, \omega\rangle + \nabla_1\Gamma^{\alpha+2}_{12}\langle e_1\wedge v_\alpha, \omega\rangle
\end{aligned}$$

and similarly

$$\begin{aligned}
\nabla_2^2 v = & -\nabla_2 h^\alpha_{21}\langle v_\alpha\wedge e_2, \omega\rangle - (h^\alpha_{21})^2\langle e_1\wedge e_2, \omega\rangle - h^\gamma_{21}C^\alpha_{2\gamma}\langle v_\alpha\wedge e_2, \omega\rangle\\
& -\nabla_2 h^\alpha_{22}\langle e_1\wedge v_\alpha, \omega\rangle - (h^\alpha_{22})^2\langle e_1\wedge e_2, \omega\rangle - h^\gamma_{22}C^\alpha_{2\gamma}\langle e_1\wedge v_\alpha, \omega\rangle\\
& +2h^\alpha_{21}h^\beta_{22}\langle v_\alpha\wedge v_\beta, \omega\rangle + \nabla_2\Gamma^{\alpha+2}_{21}\langle v_\alpha\wedge e_2, \omega\rangle + \nabla_2\Gamma^{\alpha+2}_{22}\langle e_1\wedge v_\alpha, \omega\rangle.
\end{aligned}$$

By the Coddazi–Mainardi equation

$$\begin{aligned}
\nabla_1 h^\alpha_{21} &= \nabla_2 h^\alpha_{11} + h^\gamma_{11}C^\alpha_{2\gamma} - h^\gamma_{21}C^\alpha_{1\gamma} - R_{(\alpha+2)\,121},\\
\nabla_2 h^\alpha_{12} &= \nabla_1 h^\alpha_{22} + h^\gamma_{22}C^\alpha_{1\gamma} - h^\gamma_{12}C^\alpha_{2\gamma} - R_{(\alpha+2)\,212}
\end{aligned}$$

and

$$\begin{aligned}
\langle v_1\wedge v_2, \omega\rangle &= \langle e_1\wedge e_2, \omega\rangle\\
\langle e_1\wedge v_1, \omega\rangle &= -\langle e_2\wedge v_2, \omega\rangle\\
\langle e_1\wedge v_2, \omega\rangle &= \langle e_2\wedge v_1, \omega\rangle,
\end{aligned}$$

we obtain

$$\begin{aligned}
\left(\frac{\partial}{\partial t}-\Delta\right)v = & ((h^\alpha_{11})^2 + 2(h^\alpha_{12})^2 + (h^\alpha_{22})^2 - 2h^1_{11}h^2_{12} + 2h^2_{11}h^1_{12}\\
& -2h^1_{21}h^2_{22} + 2h^2_{21}h^1_{22})\langle e_1\wedge e_2, \omega\rangle\\
& -R_{(\alpha+2)\,121}\langle e_1\wedge v_\alpha, \omega\rangle - R_{(\alpha+2)\,212}\langle v_\alpha\wedge e_2, \omega\rangle\\
& -(\nabla_1\Gamma^3_{11} - \nabla_1\Gamma^4_{12} + \nabla_2\Gamma^3_{21} - \nabla_2\Gamma^4_{22})\langle e_1\wedge v_2, \omega\rangle\\
& -(\nabla_1\Gamma^4_{11} + \nabla_1\Gamma^3_{12} + \nabla_2\Gamma^4_{21} + \nabla_2\Gamma^3_{22})\langle e_1\wedge v_1, \omega\rangle\\
= & |DJ_{\Sigma_t}|^2 v - (R_{1213}+R_{2124})\langle e_1\wedge v_1, \omega\rangle - (R_{1214}-R_{2123})\langle e_1\wedge v_2, \omega\rangle\\
& -(\nabla_1\Gamma^3_{11} - \nabla_1\Gamma^4_{12} + \nabla_2\Gamma^3_{21} - \nabla_2\Gamma^4_{22})\langle e_1\wedge v_2, \omega\rangle\\
& -(\nabla_1\Gamma^4_{11} + \nabla_1\Gamma^3_{12} + \nabla_2\Gamma^4_{21} + \nabla_2\Gamma^3_{22})\langle e_1\wedge v_1, \omega\rangle.
\end{aligned}$$

By the formula (1.16) in [Ya, Chap. IV], we get

$$H_{12} = \cos\alpha(R_{1212} + R_{1234}) + \sin\alpha(R_{1213} - R_{1224}),$$

where H_{ij} is defined in [Ya]. By the formula (1.17) in [Ya, Chapter IV], we get

$$H_{12} = R_{1s} J^s_2.$$

Since M is a Kähler–Einstein manifold, we have

$$H_{12} = -\frac{R}{2}\cos\alpha,$$

where R is the scalar curvature. By the formula (1.10) in Chapter IV in [Ya], we get

$$R_{1212} = R_{1212}\cos^2\alpha - R_{1234}\sin^2\alpha + \sin\alpha\cos\alpha(R_{1213} - R_{1224}),$$

so we obtain

$$R_{1212} + R_{1234} = \frac{\cos\alpha}{\sin\alpha}(R_{1213} - R_{1224}).$$

It follows that

$$-\frac{1}{2}R\cos\alpha = \frac{1}{\sin\alpha}(R_{1213} - R_{1224}).$$

Thus

$$R_{1213} - R_{1224} = -\tfrac{1}{2}R\sin\alpha\cos\alpha. \tag{6}$$

One checks directly that

$$\begin{aligned}
\nabla_1\Gamma^4_{11} &= -\nabla_1\Gamma^1_{41} = 0\\
\nabla_1\Gamma^3_{12} &= -\nabla_1\Gamma^1_{23} = R_{1213}\\
\nabla_2\Gamma^4_{21} &= -\nabla_2\Gamma^2_{14} = -R_{1224}\\
\nabla_2\Gamma^3_{22} &= -\nabla_2\Gamma^2_{32} = 0.
\end{aligned}$$

We therefore have

$$\left(\frac{\partial}{\partial t}-\Delta\right) v=|\bar{\nabla} J_{\Sigma_t}|^2\, v-2(R_{1213}-R_{1224})\langle e_1\wedge v_1,\, \omega\rangle.$$

By (4) and (6), we have

$$\left(\frac{\partial}{\partial t}-\Delta\right)\cos\alpha=|\bar{\nabla} J_{\Sigma_t}|^2\cos\alpha+R\sin^2\alpha\cos\alpha.$$

This proves the proposition. Q.E.D.

Then by the parabolic minimum principle, we obtain (cf. [CT2]):

THEOREM 3.3. *Let M be a Kähler–Einstein surface with nonnegative scalar curvature. Suppose that the smooth solution of the mean curvature flow exists on $[0, T)$, $T\leqslant\infty$. Set $v=\cos\alpha$ where α is the Kähler angle of Σ_t in M. If $v(x, 0)\geqslant v_0>0$, then*

$$v(x, t)\geqslant v_0,$$

$$\left(\frac{\partial}{\partial t}-\Delta\right) v\geqslant|\bar{\nabla} J_{\Sigma_t}|^2\, v$$

for all $t\in[0, T)$.

4. MONOTONICITY FORMULAS AND TYPE I SINGULARITIES

Let $H(\mathbf{X}, \mathbf{X}_0, t)$ be the backward heat kernel on R^4. Define

$$\rho(\mathbf{X}, t)=4\pi(t_0-t)\, H(\mathbf{X}, \mathbf{X}_0, t)=\frac{1}{4\pi(t_0-t)}\exp\left(-\frac{|\mathbf{X}-\mathbf{X}_0|^2}{4(t_0-t)}\right)$$

for $t<t_0$. We prove two monotonicity inequalities in this section, the first one was essentially proved by Huisken [H1] and Hamilton [Ha]. Using the second monotonicity derived in this section in our blow-up analysis at singularities, we prove that the mean curvature flow has no Type I singularity provided $v(x, 0)\geqslant v_0>0$. We first prove the monotonicity inequality on R^4. The monotonicity inequality for heat flow of harmonic maps was proved by Struwe [St] in the Euclidean case and was proved by Chen and Struwe [CS] in the general case.

PROPOSITION 4.1. *Let $M^4 = R^4$. We have*

$$\frac{\partial}{\partial t}\int_{\Sigma_t} \rho(F, t)\, d\mu_t = -\int_{\Sigma_t} \rho(F, t)\left|\mathbf{H}+\frac{(F-\mathbf{X}_0)^\perp}{2(t_0-t)}\right|^2 d\mu_t.$$

Proof. It is clear that

$$\begin{aligned}\frac{\partial}{\partial t}\int_{\Sigma_t} \rho(F, t)\, d\mu_t &= \int_{\Sigma_t}\frac{\partial}{\partial t}\rho(F, t)\, d\mu_t - \int_{\Sigma_t}\rho(F, t)\,|\mathbf{H}|^2\, d\mu_t \\ &= \int_{\Sigma_t}\left(\frac{\partial}{\partial t}+\Delta\right)\rho(F, t)\, d\mu_t - \int_{\Sigma_t}\rho(F, t)\,|\mathbf{H}|^2\, d\mu_t.\end{aligned}$$

Straight computation leads to

$$\frac{\partial}{\partial t}\rho(\mathbf{X}, t) = \left(\frac{1}{t_0-t}-\frac{1}{2(t_0-t)}\langle \mathbf{H}, \mathbf{X}-\mathbf{X}_0\rangle - \frac{|\mathbf{X}-\mathbf{X}_0|^2}{4(t_0-t)^2}\right)\rho(\mathbf{X}, t)$$

and

$$\nabla \exp\left(-\frac{|\mathbf{X}-\mathbf{X}_0|^2}{4(t_0-t)}\right) = -\exp = \left(-\frac{|\mathbf{X}-\mathbf{X}_0|^2}{4(t_0-t)}\right)\frac{\langle \mathbf{X}-\mathbf{X}_0, \nabla\mathbf{X}\rangle}{2(t_0-t)}$$

and

$$\begin{aligned}&\Delta \exp\left(-\frac{|\mathbf{X}-\mathbf{X}_0|^2}{4(t_0-t)}\right) \\ &= \exp\left(-\frac{|\mathbf{X}-\mathbf{X}_0|^2}{4(t_0-t)}\right)\left(\frac{|\langle \mathbf{X}-\mathbf{X}_0, \nabla\mathbf{X}\rangle|^2}{4(t_0-t)^2} - \frac{\langle \mathbf{X}-\mathbf{X}_0, \Delta\mathbf{X}\rangle}{2(t_0-t)} - \frac{|\nabla\mathbf{X}|^2}{2(t_0-t)}\right).\end{aligned}$$

Note that in the induced metric on Σ_t

$$|\nabla F|^2 = 2 \qquad \text{and} \qquad \Delta F = \mathbf{H},$$

so we have

$$\left(\frac{\partial}{\partial t}+\Delta\right)\rho(F, t) = -\left(\frac{\langle F-\mathbf{X}_0, \mathbf{H}\rangle}{(t_0-t)} + \frac{|(F-\mathbf{X}_0)^\perp|^2}{4(t_0-t)^2}\right)\rho(F, t) \tag{7}$$

Then the proposition follows. Q.E.D.

We then prove the monotonicity inequality on a curved manifold M^4. Let i_M be the injective radius of M^4. We choose a cut off function $\phi \in C_0^\infty(B_{2r}(\mathbf{X}_0))$ with $\phi \equiv 1$ in $B_r(\mathbf{X}_0)$, where $\mathbf{X}_0 \in M$, $0 < 2r < i_M$. Choose a normal coordinates in $B_{2r}(\mathbf{X}_0)$ and express F using the coordinates (F^1, F^2, F^3, F^4) as a surface in R^4. We define

$$\Phi(\mathbf{X}_0, t_0, t) = \int_{\Sigma_t} \phi(F)\, \rho(F, t)\, d\mu_t.$$

Hamilton used the backward heat kernel on M^4 in his monotonicity in [Ha]. We use the backward heat kernel on R^4 to localize near a point. For harmonic map heat flow, similar localization was taken in [CS] on the domain rather then on the target manifold.

PROPOSITION 4.2. *Let M^4 be a compact Riemannian 4-manifold. There are positive constants c_1 and c_2 depending only on M^4, F_0 and r which is the constant in the definition of ϕ, such that*

$$\frac{\partial}{\partial t} e^{c_1 \sqrt{t_0 - t}} \Phi(\mathbf{X}_0, t_0, t) \leqslant -e^{c_1 \sqrt{t_0 - t}} \int_{\Sigma_t} \phi \rho(F, t) \left| \mathbf{H} + \frac{(F - \mathbf{X}_0)^\perp}{2(t_0 - t)} \right|^2 d\mu_t + c_2 (t_0 - t).$$

Proof. In this case, we have

$$\begin{aligned} \frac{\partial}{\partial t} \Phi(\mathbf{X}_0, t_0, t) &= \int_{\Sigma_t} \left(\frac{\partial}{\partial t} + \Delta \right) \phi \rho(F, t)\, d\mu_t - \int_{\Sigma_t} \phi \rho(F, t)\, |\mathbf{H}|^2\, d\mu_t \\ &= \int_{\Sigma_t} \phi \left(\frac{\partial}{\partial t} + \Delta \right) \rho(F, t)\, d\mu_t - \int_{\Sigma_t} \phi \rho(F, t)\, |\mathbf{H}|^2\, d\mu_t \\ &\quad + \int_{\Sigma_t} \Delta \phi \rho(F, t)\, d\mu_t + 2 \int_{\Sigma_t} \nabla \phi \cdot \nabla \rho(F, t)\, d\mu_t. \end{aligned}$$

Notice that

$$\Delta F = \mathbf{H} + g^{ij} \Gamma_{ij}^{\alpha} v_\alpha$$

and we then have

$$\begin{aligned} &\left(\frac{\partial}{\partial t} + \Delta \right) \rho(F, t) \\ &\quad = -\left(\frac{\langle F - \mathbf{X}_0, \mathbf{H} \rangle}{(t_0 - t)} + \frac{|(F - \mathbf{X}_0)^\perp|^2}{4(t_0 - t)^2} + \frac{\langle F - \mathbf{X}_0, g^{ij} \Gamma_{ij}^{\alpha} v_\alpha \rangle}{(t_0 - t)} \right) \rho(F, t). \qquad (8) \end{aligned}$$

Note that $\Delta\phi = 0$, $\nabla\phi = 0$ in $B_r(\mathbf{X}_0)$, we can see that

$$|\Delta\phi\rho(F,t)| \leqslant C \qquad \text{and} \qquad |\nabla\phi\nabla\rho(F,t)| \leqslant C,$$

where the constant C depends on r and M, hence

$$\int_{\Sigma_t} \Delta\phi\rho(F,t)\, d\mu_t \leqslant C \int_{\Sigma_t} d\mu_t \leqslant C$$

$$\int_{\Sigma_t} \nabla\phi \cdot \nabla\rho(F,t)\, d\mu_t \leqslant C \int_{\Sigma_t} d\mu_t \leqslant C.$$

Since $\Gamma^{\alpha}_{ij}(\mathbf{X}_0) = 0$, we have $|g^{ij}\Gamma^{\alpha}_{ij}(F)| \leqslant C\,|F - \mathbf{X}_0|$. thus

$$\frac{\langle F - \mathbf{X}_0, g^{ij}\Gamma^{\alpha}_{ij} v_\alpha \rangle}{(t_0 - t)} \leqslant C\, \frac{|F - \mathbf{X}_0|^2}{t_0 - t}.$$

Hence

$$\frac{\langle F - \mathbf{X}_0, g^{ij}\Gamma^{\alpha}_{ij} v_\alpha \rangle}{(t_0 - t)}\, \rho(F,t) \leqslant c_1\, \frac{\rho(F,t)}{\sqrt{t_0 - t}} + C.$$

It concludes that

$$\frac{\partial}{\partial t}\Phi(\mathbf{X}_0, t_0, t) \leqslant -\int_{\Sigma_t} \phi\rho(F,t) \left| \mathbf{H} + \frac{(F - \mathbf{X}_0)^{\perp}}{2(t_0 - t)} \right|^2 d\mu_t + \frac{c_1}{\sqrt{t_0 - t}}\, \Phi(\mathbf{X}_0, t_0, t) + c_2.$$

The proposition follows. Q.E.D.

By an argument similar to the one used in the proof of Proposition 4.2, we can prove the following monotonicity inequality.

THEOREM 4.3. *Let M be a Kähler–Einstein surface with nonnegative scalar curvature. Let α be the Kähler angle of the surface Σ_t in M. Suppose that* $\cos\alpha(\cdot, 0)$ *has a positive lower bound. Let* $v = \cos\alpha$. *Then we have*

$$\begin{aligned}
&\frac{\partial}{\partial t} e^{c_1\sqrt{t_0 - t}} \int_{\Sigma_t} \frac{1}{v}\, \phi\rho(F,t)\, d\mu_t \\
&\quad\leqslant -e^{c_1\sqrt{t_0 - t}} \left(\int_{\Sigma_t} \frac{1}{v}\, \phi\rho(F,t) \left| \mathbf{H} + \frac{(F - \mathbf{X}_0)^{\perp}}{2(t_0 - t)} \right|^2 d\mu_t \right. \\
&\quad\quad \left. + \int_{\Sigma_t} \frac{1}{v}\, \phi\rho(F,t)\, |\overline{\nabla} J_{\Sigma_t}|^2\, d\mu_t + \int_{\Sigma_t} \frac{2}{v^3}\, |\nabla v|^2\, \phi\rho(F,t)\, d\mu_t \right) \\
&\quad\quad + c_2(t_0 - t).
\end{aligned}$$

Here the positive constants c_1 and c_2 depend on M, F_0 and r which is the constant in the definition of ϕ.

Proof. By Theorem 3.3, we have

$$\left(\frac{\partial}{\partial t}-\Delta\right)\frac{1}{v}\leqslant -|\bar{\nabla} J_{\Sigma_t}|^2\frac{1}{v}-\frac{2}{v^3}|\nabla v|^2.$$

So,

$$\begin{aligned}\frac{\partial}{\partial t}\int_{\Sigma_t}\frac{1}{v}\phi\rho(F,t)\,d\mu_t &= \int_{\Sigma_t}\frac{\partial}{\partial t}\frac{1}{v}\phi\rho(F,t)\,d\mu_t+\int_{\Sigma_t}\frac{\partial}{\partial t}\rho(F,t)\,\phi\frac{1}{v}\,d\mu_t\\ &\quad -\int_{\Sigma_t}\rho(F,t)\,\phi\,|\mathbf{H}|^2\,d\mu_t\\ &\leqslant \int_{\Sigma_t}\phi\left(\frac{\partial}{\partial t}+\Delta\right)\rho(F,t)\frac{1}{v}\,d\mu_t-\int_{\Sigma_t}\rho(F,t)\,\phi\,|\mathbf{H}|^2\,d\mu_t\\ &\quad -\int_{\Sigma_t}|\bar{\nabla} J_{\Sigma_t}|^2\frac{1}{v}\phi\rho(F,t)\,d\mu_t-\int_{\Sigma_t}\phi\frac{2}{v^3}|\nabla v|^2\rho(F,t)\,d\mu_t\\ &\quad +\int_{\Sigma_t}\Delta\phi\frac{1}{v}\rho(F,t)\,d\mu_t+2\int_{\Sigma_t}\nabla\phi\cdot\nabla\rho(F,t)\frac{1}{v}\,d\mu_t.\end{aligned}$$

Using (8) and the estimates in the proof of Proposition 4.2, we obtain the monotonicity inequality. Q.E.D.

The standard parabolic theory implies that (1) has a smooth solution for short time. We state it in the following theorem.

THEOREM 4.4. *There exists $T>0$ such that* (1) *has a smooth solution in the time interval* $[0,T)$. *If* $\max_{\Sigma_t}|\mathbf{A}|^2$ *is bounded near T, the solution can be extended to* $[0,T+\epsilon)$ *for some* $\epsilon>0$.

However, in general $\max_{\Sigma_t}|\mathbf{A}|^2$ becomes unbounded as $t\to T$.

DEFINITION 4.5. We say that the mean curvature flow F has Type I singularity at $T>0$, if

$$\varlimsup_{t\to T}\,(T-t)\max_{\Sigma_t}|\mathbf{A}|^2\leqslant C,$$

for some positive constant C.

LEMMA 4.6. *Let $U(t) = \max_{\Sigma_t} |\mathbf{A}|^2$. If the mean curvature flow blows up at $T > 0$, there is a positive constant c depending only on M^4, such that, if $0 < T - t < \pi/16\sqrt{c}$, the function $U(t)$ satisfies*

$$U(t) \geqslant \frac{1}{4\sqrt{2}\,(T-t)}.$$

Proof. By Proposition 2.6 and the parabolic maximum principle, we have

$$\begin{aligned}\frac{\partial}{\partial t} U(t) &\leqslant 2(U(t))^2 + c_1 U(t) + c_2\sqrt{U(t)} \\ &\leqslant 4(U(t))^2 + 4c,\end{aligned}$$

where c_1 and c_2 are constants which only depend on $\max |R|$, $\max |\nabla R|$ for the Riemannian curvature R on M and the constant c depends on c_1, c_2. so,

$$\frac{\partial}{\partial t}\left(\arctan\left(\frac{\sqrt{c}}{U(t)}\right)\right) \geqslant -4\sqrt{c}\,. \tag{9}$$

Since the mean curvature flow blows up at $T > 0$, we have $U(t) \to \infty$ as $t \to T$. Integrating the last inequality from t to T, we get, if $0 < T - t < \frac{\pi}{16\sqrt{c}}$,

$$\frac{\sqrt{c}}{U(t)} \leqslant \tan(4\sqrt{c}\,(T-t)) \leqslant \sqrt{2}\cdot 4\sqrt{c}\,(T-t).$$

This implies the desired inequality. Q.E.D.

THEOREM 4.7. *Let M^4 be a Kähler–Einstein surface with nonnegative scalar curvature. If the Kähler angle of the initial surface $v(x, 0) \geqslant v_0 > 0$, then the mean curvature flow* (1) *has no Type I singularity at any $T > 0$.*

Proof. Suppose that the mean curvature flow has a Type I singularity at $t_0 > 0$. Assume that

$$\lambda_k^2 = |\mathbf{A}|^2\,(x_k, t_k) = \max_{t \leqslant t_k} |\mathbf{A}|^2$$

and $x_k \to p \in \Sigma$, $t_k \to T$ as $k \to \infty$. We choose a local coordinate system on M^4 such that $F(p, t_0) = 0$. We rescale the mean curvature flow at 0, more precisely, we set

$$F_k(x, t) = \lambda_k(F(x, \lambda_k^{-2} t + t_k) - F(p, t_k)).$$

$d_{g_k}(X_k, p) <$

$\lambda_k d_{g(t_k)}(x_k, p) \leqslant C$

Denote by Σ_t^k the scaled surface $F_k(\cdot, t)$ and take

$$g_{ij}^k = \lambda_k^2 g_{ij}, \; (g^k)^{ij} = \lambda_k^{-2} g^{ij}.$$

We therefore have

$$\frac{\partial F_k}{\partial t} = \lambda_k^{-1} \frac{\partial F}{\partial t} \tag{10}$$

$$\Delta_{g^k} F_k = \lambda_k^{-1} \Delta F \tag{11}$$

and

$$\frac{\partial F_k}{\partial t} = \mathbf{H}_k \tag{12}$$

$$|\mathbf{A}_k|^2 = \lambda_k^{-2} |\mathbf{A}|^2. \tag{13}$$

By Lemma 4.6, we have

$$\frac{c}{(t_0 - t_k)} \leqslant |\mathbf{A}|^2 (x, t_k) \leqslant \frac{C}{(t_0 - t_k)}$$

for some uniform constants c and C independent of k, so

$$|\mathbf{A}_k|^2 \leqslant 1 \qquad \text{and} \qquad |\mathbf{A}_k|^2 (p, 0) \geqslant c > 0.$$

There is a subsequence of F_k which we also denote by F_k, such that $F_k \to F_\infty$ as $k \to \infty$ in any ball $B_R(0) \subset R^4$, and F_∞ satisfies

$$\frac{\partial F_\infty}{\partial t} = \mathbf{H}_\infty \qquad \text{with} \quad |\mathbf{A}_\infty(0, 0)| \geqslant c > 0. \tag{14}$$

Recall that Σ_t is the curve defined by $F(\cdot, t)$. For any $R > 0$, we choose a cut-off function $\phi_R \in C_0^\infty(B_{2R}(0))$ with $\phi_R \equiv 1$ in $B_R(0)$ where $B_r(0)$ is the metric ball centered at 0 with radius r. It is clear that, for k sufficiently large,

$$\int_{\Sigma_t^k} \frac{1}{v} \frac{1}{0-t} \phi_R(F_k) \exp\left(-\frac{|F_k + \lambda_k F(p, t_k)|^2}{4(0-t)}\right) d\mu_t^k$$
$$= \int_{\Sigma_{t_k + \lambda_k^{-2} t}} \frac{1}{v} \phi_R(F) \frac{1}{t_k - (t_k + \lambda_k^{-2} t)} \exp\left(-\frac{|F(x, t_k + \lambda_k^{-2} t)|^2}{4(t_k - (t_k + \lambda_k^{-2} t))}\right) d\mu_t,$$

where ϕ is the function defined in Theorem 4.3. Note that $t_k+\lambda_k^{-2}t\to t_0$ for any fixed t. Using Theorem 4.3 by taking the initial time to be t_k, we see that for any $-\infty<s_1<s_2<0$,

$$
\begin{aligned}
& e^{c_1\sqrt{t_k-(t_k+\lambda_k^{-2}s_2)}}\int_{\Sigma^k_{s_2}}\frac{1}{v}\phi_R\frac{1}{0-s_2}\exp\left(-\frac{|F_k+\lambda_kF(p,t_k)|^2}{4(0-s_2)}\right)d\mu^k_{s_2}\\
&\quad -e^{c_1\sqrt{t_k-(t_k+\lambda_k^{-2}s_1)}}\int_{\Sigma^k_{t_1}}\frac{1}{v}\phi_R\frac{1}{0-s_1}\exp\left(-\frac{|F_k+\lambda_kF(p,t_k)|^2}{4(0-s_1)}\right)d\mu^k_{s_1}\\
&\quad \to 0 \qquad \text{as} \quad k\to\infty.
\end{aligned}
$$

By Theorem 4.3, if we take the initial time to be 0, then for any $-\infty<s_1<s_2<0$,

$$
\begin{aligned}
& -e^{c_1\sqrt{-\lambda_k^{-2}s_2}}\int_{\Sigma^k_{s_2}}\frac{1}{v}\phi_R\frac{1}{0-s_2}\exp\left(-\frac{|F_k+\lambda_kF(p,t_k)|^2}{4(0-s_2)}\right)d\mu^k_{s_2}\\
&\quad +e^{c_1\sqrt{-\lambda_k^{-2}s_1}}\int_{\Sigma^k_{s_1}}\frac{1}{v}\phi_R\frac{1}{0-s_1}\exp\left(-\frac{|F_k+\lambda_kF(p,t_k)|^2}{4(0-s_1)}\right)d\mu^k_{s_1}\\
&\quad \geqslant\int_{s_1}^{s_2}e^{c_1\sqrt{-\lambda_k^{-2}t}}\int_{\Sigma^k_t}\frac{1}{v}\phi_R\rho(F_k,t)\left|\mathbf{H}_k+\frac{(F_k+\lambda_kF(p,t_k))^\perp}{2(t_0-t)}\right|^2d\mu_t\\
&\quad +\int_{s_1}^{s_2}e^{c_1\sqrt{-\lambda_k^{-2}t}}\int_{\Sigma^k_t}\frac{1}{v}\phi_R\rho(F_k,t)\,|\overline{\nabla}J_{\Sigma_{t_k}}|^2\,d\mu^k_t\\
&\quad +\int_{s_1}^{s_2}e^{c_1\sqrt{-\lambda_k^{-2}t}}\int_{\Sigma^k_t}\frac{2}{v^3}|\nabla v|^2\,\phi_R\rho(F_k,t)\,d\mu^k_t\\
&\quad -c_2\lambda_k^{-1}(s_2-s_1).
\end{aligned}
\tag{15}
$$

Since

∂F

$$
|F(p,t_k)|\leqslant\int_{t_k}^{t_0}\left|\frac{\partial F_k}{\partial t}\right|dt\leqslant\int_{t_k}^{t_0}|\mathbf{H}|\,dt\leqslant C\sqrt{(t_0-t_k)}\leqslant\frac{C}{\lambda_k},
$$

we assume that $\lambda_kF(p,t_k)\to q$ as $k\to\infty$. Letting $k\to\infty$, by (15), we get

$$DJ_\infty\equiv0,$$

where D is the derivatives in R^4. By Lemma 3.1, we can see that

$$\mathbf{H}_\infty\equiv0$$

and hence

$$\frac{\partial F_\infty}{\partial t} \equiv 0.$$

Applying (15) again, we get

$$(F_\infty + q)^\perp \equiv 0.$$

That is,

$$\langle F_\infty + q, v_\alpha \rangle = 0.$$

Note that the above inner product is taken in R^4, and differentiating in R^4 then yields

$$0 = \langle \partial_i F_\infty, v_\alpha \rangle + \langle F_\infty + q, \partial_i v_\alpha \rangle = \langle F_\infty + q, \partial_i v_\alpha \rangle$$

because $\partial_i F_\infty$ is tangential to Σ_∞ and then by the Weingarten equation we have

$$(h_\infty)^\alpha_{ij} \langle F_\infty + q, e_j \rangle = 0 \qquad \text{for all} \quad \alpha, i = 1, 2.$$

So for $\alpha = 1, 2$, we have

$$\det((h_\infty)^\alpha_{ij}) = 0.$$

Since $\mathbf{H} = 0$, we also have for $\alpha = 1, 2$ we also have

$$tr((h_\infty)^\alpha_{ij}) = 0.$$

These result in that the symmetric matrix $((h_\infty)^\alpha_{ij})$ is the zero matrix. In other words, we have $(h_\infty)^\alpha_{ij} = 0$ for all $i, j, \alpha = 1, 2$, which yields that $|\mathbf{A}_\infty| \equiv 0$. However, (14) asserts $|\mathbf{A}_\infty| \not\equiv 0$, and hence we get a contradiction. Q.E.D.

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